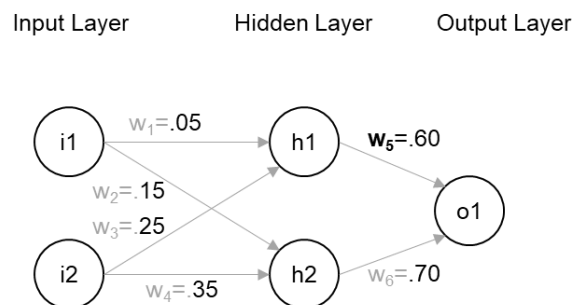


Exercise 1: Neural Networks

Consider the following neural network with two input nodes, two nodes in a single hidden layer, and one output node. The network's current weights are indicated on the edges.



What we are doing. We walk through one training step by hand. A forward pass (parts a and b) sends the inputs through the network to produce its output and the resulting error. One backward pass (part c) then uses that error to update a single weight. Repeating this over all weights, many times, is how a network learns.

Notation. i = input layer (nodes i1, i2), h = hidden layer (h1, h2), o = output layer (o1). For each node, z is the net input (weighted sum of incoming values) and $\Phi(z)$ is its output after the sigmoid. The weights w1 to w6 sit on the edges. Sidenote: all bias units are assumed to be zero in this exercise.

- a) For **input values of i1 = 1, i2 = 2**, compute the nodes' output values $\Phi(z)$ as well as the intermediate z values. Please use the sigmoid activation function (see below). Assume all bias units (b) to be zero.

We begin by calculating the intermediate z-value for node h1:

$$z_1 = w_1 * i_1 + w_3 * i_2 = 0,05 * 1 + 0,25 * 2 = \mathbf{0,55}$$

Next, we transform the intermediate value into the nodes' output value by applying the sigmoid activation function:

$$\Phi(z_{h1}) = \frac{1}{1 + e^{-0,55}} = \frac{1}{1 + 0,577} = \mathbf{0,634}$$

We repeat both steps for node h2:

$$z_{h2} = w_2 * i_1 + w_4 * i_2 = 0,15 * 1 + 0,35 * 2 = \mathbf{0,85}$$

$$\Phi(z_{h2}) = \frac{1}{1 + e^{-0,85}} = \frac{1}{1+0,427} = \mathbf{0,701}$$

We use the calculated output values of h1 and h2 to calculate the intermediate z-value of o1:

$$z_{o1} = w_5 * f(z_1) + w_6 * f(z_2) = 0,6 * 0,634 + 0,7 * 0,701 = \mathbf{0,871}$$

Lastly, we plug the intermediate value of o1 into the sigmoid activation function to arrive at the final output value:

$$\Phi(z_{o1}) = \frac{1}{1 + e^{-0,871}} = \mathbf{0,705}$$

	z (=net)	Φ(z) (=out)
h1	0,55	0,634
h2	0,85	0,701
o1	0,871	0,705

b) Assume that the **target value was 0**. What is the total error?

The formula for the total error is:

$$E_{total} = \sum_i \frac{1}{2} * (target_i - output_i)^2$$

As we only have one output node o1, this yields the following total error formula:

$$E_{o1} = \frac{1}{2} * (target_{o1} - output_{o1})^2$$

By plugging in 0 for the target value and the actual output of o1 we can calculate the total error:

$$E_{total} = \frac{1}{2} * (0 - 0,705)^2 = \frac{1}{2} * (-0,705)^2 = \frac{1}{2} * 0,497 = \mathbf{0,248}$$

- c) Now, conduct one update step for w_5 (using backward propagation). To do so, make use of the piecewise multiplicativity between total and partial derivatives and first determine the individual derivatives and values.

$$\frac{\partial E_{total}}{\partial w_5} = \frac{\partial E_{total}}{\partial out_{o1}} * \frac{\partial out_{o1}}{\partial net_{o1}} * \frac{\partial net_{o1}}{\partial w_5}$$

As a reminder, the total error is given by: $E_{o1} = \frac{1}{2} * (target_{o1} - output_{o1})^2$

First, we calculate the leftmost segment of the formula by applying the chain rule and plug in our values:

$$\frac{\partial E_{total}}{\partial out_{o1}} = 2 * \frac{1}{2} (target_{o1} - output_{o1})^{2-1} * (-1) = -(\mathbf{target_{o1} - output_{o1}})$$

$$\frac{\partial E_{total}}{\partial out_{o1}} = -(0 - 0,705) = \mathbf{0,705}$$

Next, we calculate the middle piece as a formula and then plug in the values. As a reminder, the value for the output of o1 is:

$$output_{o1} = \frac{1}{1 + e^{-net_{o1}}} = \Phi(\mathbf{net_{o1}})$$

We use the given hint to calculate the derivative towards the intermediate value of o1:

$$\frac{\partial output_{o1}}{\partial net_{o1}} = \Phi(net_{o1}) * (1 - \Phi(net_{o1})) = \mathbf{output_{o1} * (1 - output_{o1})} = 0,705 * (1 - 0,705) = \mathbf{0,208}$$

Next, we calculate the rightmost and last part of our formula. As a reminder, the formula for intermediate net value is:

$$\mathbf{z_{o1}} = w_5 * f(z_{h1}) + w_6 * f(z_{h2}) + b$$

We assumed there is no bias, so $b = 0$. By calculating the derivative towards our weight w_5 , we receive:

$$\frac{\partial net}{\partial w_5} = \mathbf{f(z_{h1}) = 0,634}$$

Next, we plug everything together and perform the update step:

$$\begin{aligned} \frac{\partial E_{total}}{\partial w_5} &= \frac{\partial E_{total}}{\partial output_{o1}} * \frac{\partial output_{o1}}{\partial net_{o1}} * \frac{\partial net_{o1}}{\partial w_5} \\ &= -(\mathbf{target_{o1} - output_{o1}}) * \mathbf{output_{o1}} * (1 - \mathbf{output_{o1}}) * \mathbf{output_{h1}} \\ &= 0,705 * 0,208 * 0,634 = \mathbf{0,093} \end{aligned}$$

Partial Derivative	Formula	Value
$\frac{\partial E_{total}}{\partial out_{o1}}$	$-(target_{o1} - output_{o1})$	0,705
$\frac{\partial out_{o1}}{\partial net_{o1}}$	$output_{o1} * (1 - output_{o1})$	0,208
$\frac{\partial net_{o1}}{\partial w_5}$	$output_{h1}$	0,634

- d) Now, using gradient descent, conduct the update step. Assume a learning rate of 0.5. What is the new value of w_5 ?

As a reminder, the current value of w_5 is 0,6. We apply the formula for updating the factor while considering the learning rate η (eta) of 0.5:

$$w'_5 = w_5 - \eta * \frac{\partial E_{Total}}{\partial w_5} = 0,6 - 0,5 * 0,093 = \mathbf{0,554}$$

Hints:

$z = \sum_i w_i x_i + b$	Derivate of Sigmoid function: $f(x) = \frac{1}{1 + e^{-x}} = \frac{e^x}{1 + e^x}$ $\frac{d}{dx} f(x) = \frac{e^x \cdot (1 + e^x) - e^x \cdot e^x}{(1 + e^x)^2}$ $\frac{d}{dx} f(x) = \frac{e^x}{(1 + e^x)^2} = f(x)(1 - f(x))$
$\phi(z) = \frac{1}{1 + e^{-z}}$	
$E_{total} = \sum \frac{1}{2} (target - output)^2$	

Exercise 2: Image Classification

Consider the datasets furniture images `scandi.zip`, `modern.zip`, and `pop.zip` (as provided through ISIS). Make yourself familiar with the online service <https://app.ximilar.com> and use it to train a model that distinguishes between the furniture styles. Which accuracy can you reach?

Example images

Scandi



Modern



Pop



The exercise was discussed in the tutorial, feel free to familiarize yourself with the platform and available tools further.